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## Quiz 1 – Introduction to OS

| Q1  | Q2  | Total |
|-----|-----|-------|
| (5) | (5) | (10)  |

L1: Student Name: \_\_\_\_\_

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- Q1) Print a string (a note) as many times as you want. This will require some loops. For instance, the program titled “task1” prints the given note three times:

(5)

```
task1 3 "I hate you"
I hate you
I hate you
I hate you
```

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
    // Need two arguments: <count> and <note>
    if (argc != 3)
    {
        printf("Usage: %s <Count-how many times> and <String - note> \n", argv[0]);
        return 1;
    }

    int count = atoi(argv[1]);

    if (count <= 0)
    {
        printf("Error: count must be a positive number.\n");
        return 1;
    }

    for (int i = 0; i < count; i++) {
        printf("%s\n", argv[2]);
    }

    return 0;
}
```

Q2) Write a program (myhead.c) that prints the first 'n' lines of a file.

(5)

```
myhead 4 file.txt // Reads the first four lines of the file, file.txt
```

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
    // Expect exactly: <program_name> <n> <filename>
    if (argc != 3)
    {
        printf("Usage: %s <n> <filename>\n", argv[0]);
        return 1;
    }

    int n = atoi(argv[1]);

    // Open the file for reading
    FILE *fp = fopen(argv[2], "r");
    if (fp == NULL) {
        printf("Could not open file: %s\n", argv[2]);
        return 1;
    }

    char line[100];
    int count = 0;

    // Keep reading lines until we've printed n of them, or we reach the end (NULL).
    while (count < n && fgets(line, sizeof(line), fp) != NULL)
    {
        printf("%s", line);
        count++;
    }

    fclose(fp);
    return 0;
}
```